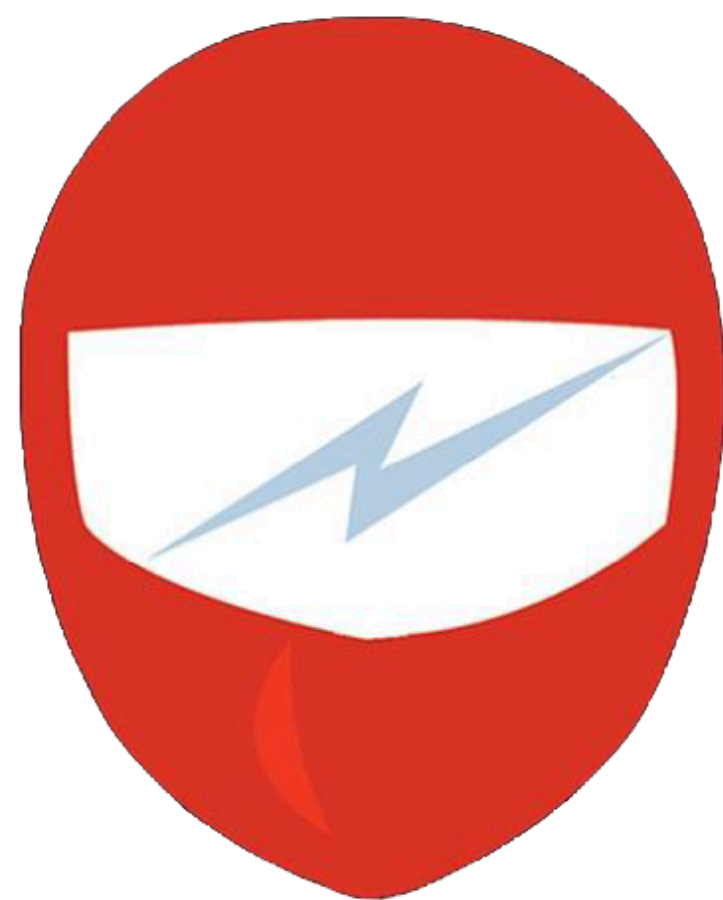




SLAM

Ahmed Shaboury

**Technical Director
Autonomous Team**



Today's Content

- ▶ What is SLAM ?
- ▶ The SLAM Problem
- ▶ SLAM Applications
- ▶ SLAM Problem
- ▶ Types of SLAM
- ▶ Sensors



What is SLAM ?



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What is Robot Mapping?



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What is Robot Mapping?

Robot – A device, that moves through the environment



What is Robot Mapping?

Robot – A device, that moves through the environment

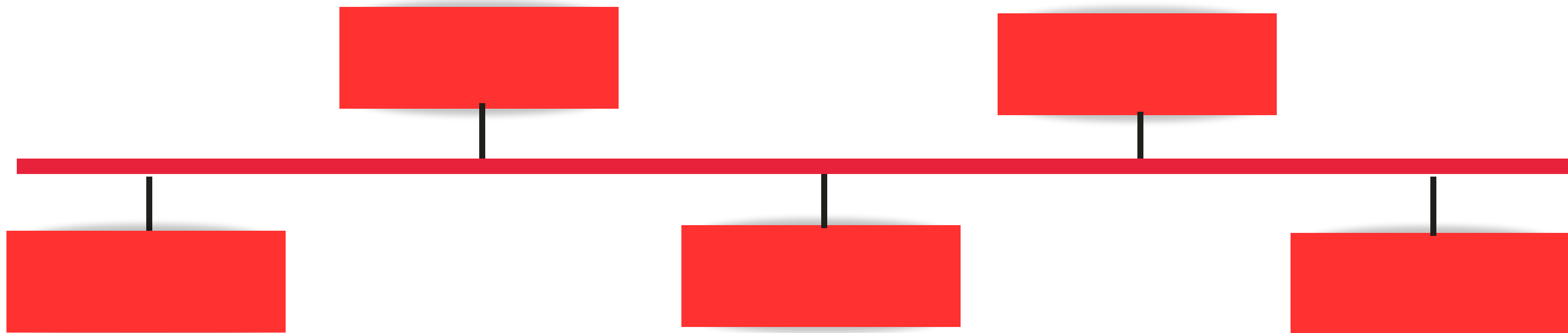
Mapping – Modelling the environment





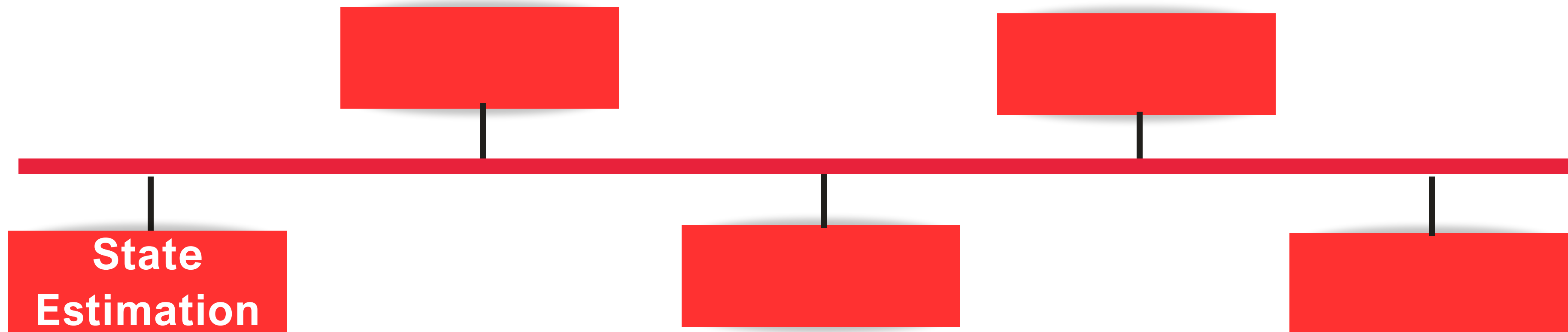
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Related Terms



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Related Terms



Related Terms

Localization

State
Estimation



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Related Terms

Localization

State
Estimation

Mapping



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Related Terms

Localization

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Localization

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Mapping

Navigation



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What is SLAM ?

- Computing the robot's poses and the map of the environment at the same time
- **Localization:** estimating the robot's location
- **Mapping:** building a map
- **SLAM:** building a map & localizing the robot simultaneously



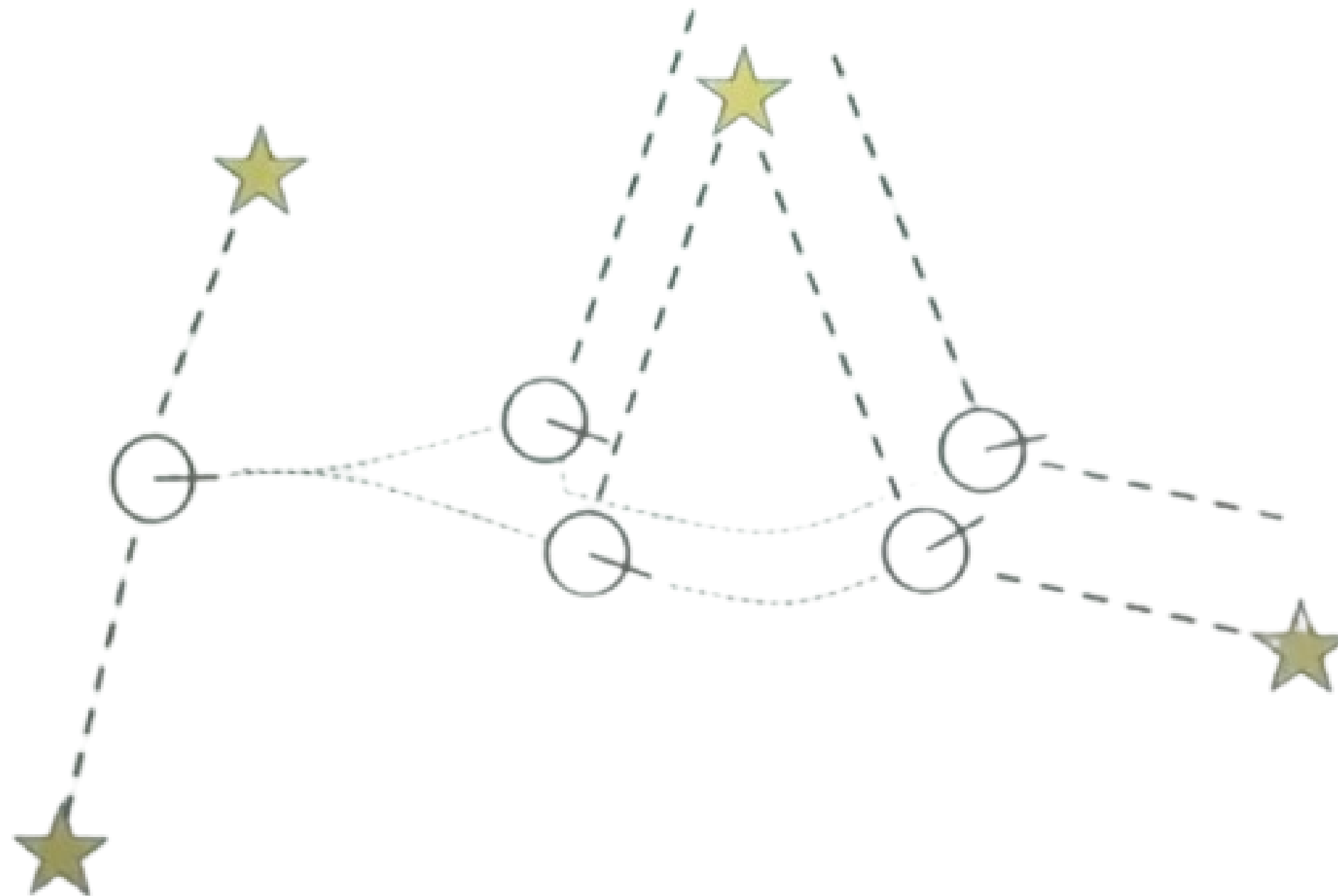
Localiztion Example:

- Estimate the robot's poses given landmarks



Localiztion Example:

- Estimate the robot's poses given landmarks



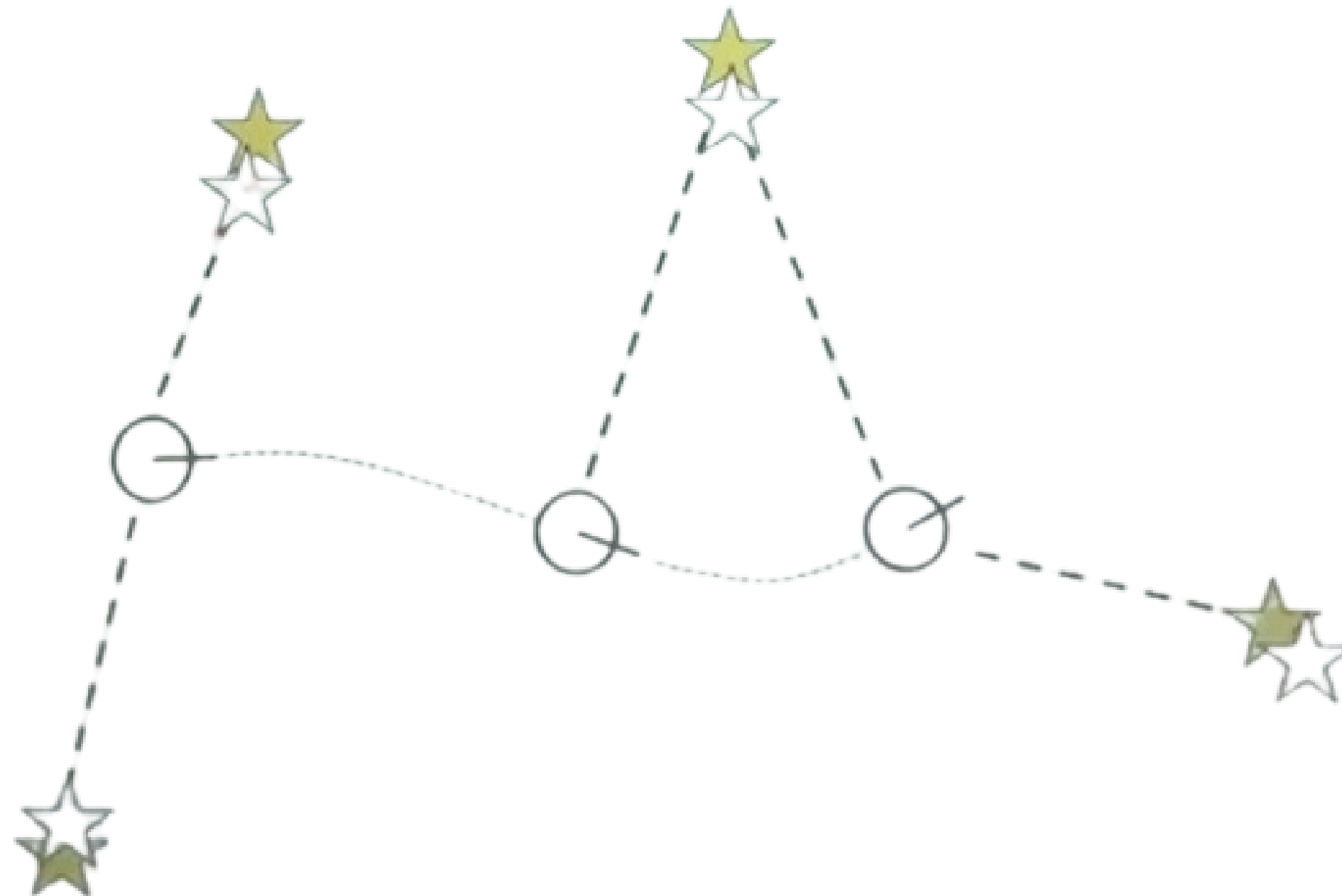
Mapping Example:

- Estimate the landmarks given the robot's poses



Mapping Example:

- Estimate the landmarks given the robot's poses



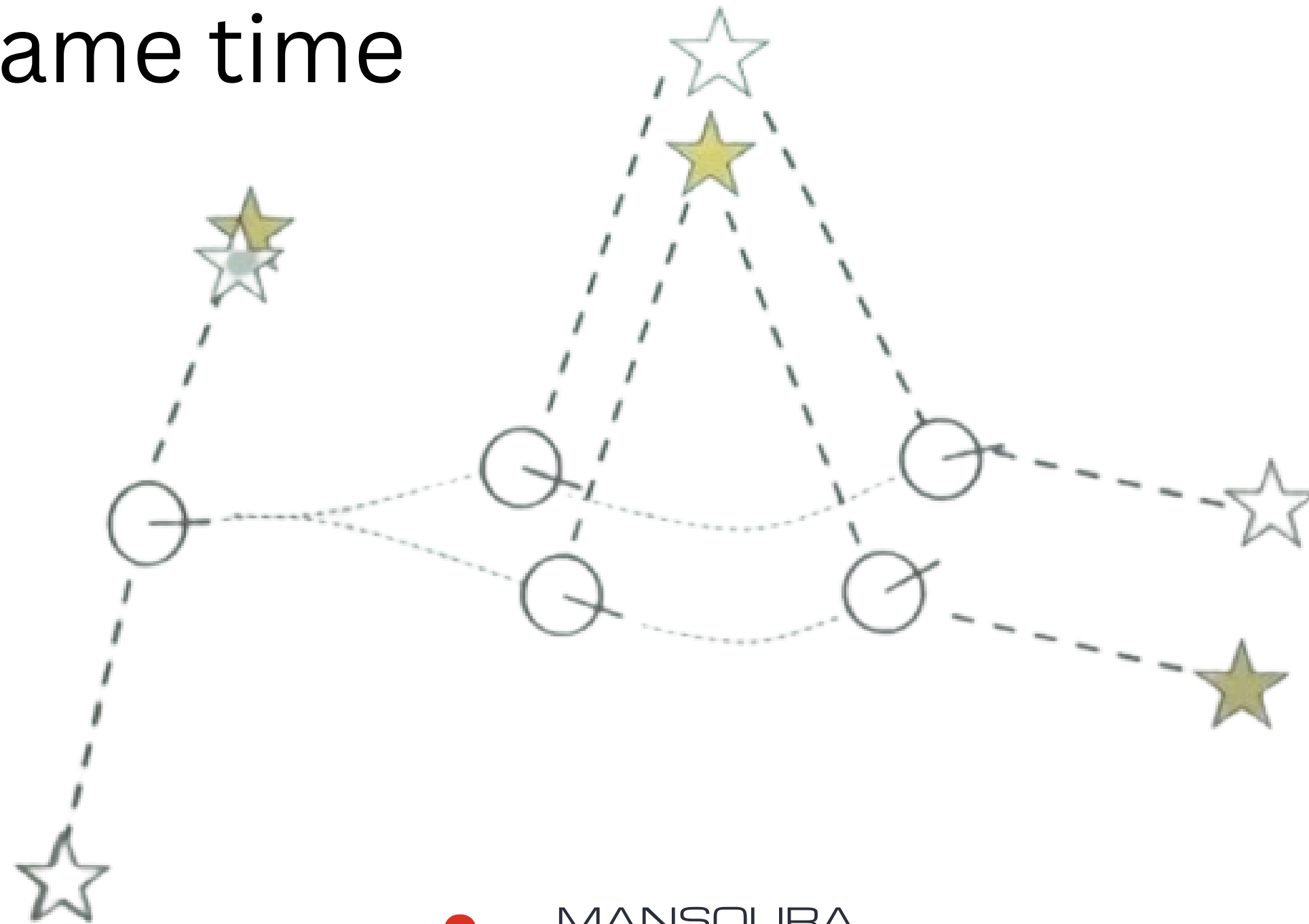
SLAM Example:

- Estimate the robot's poses and the landmarks at the same time



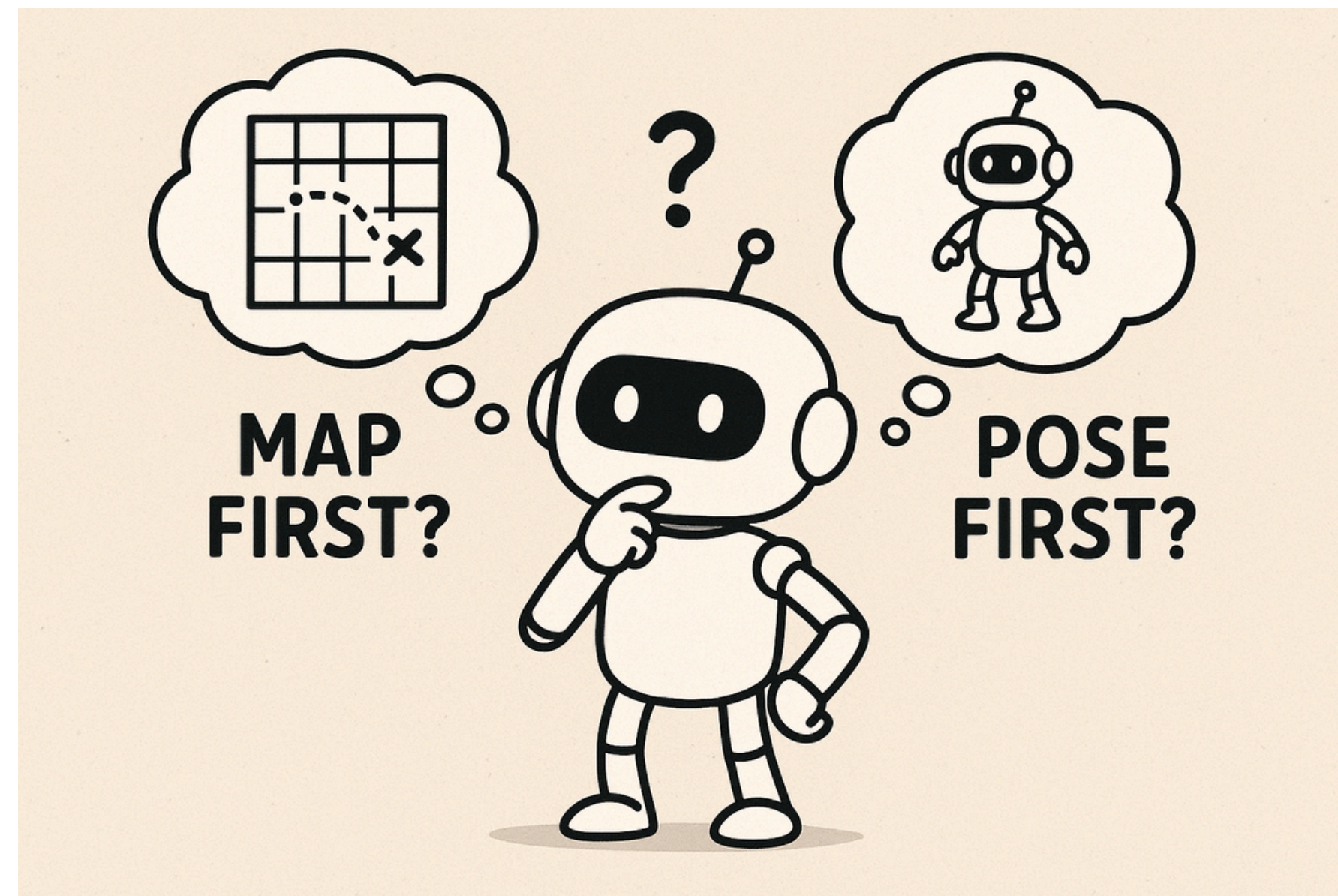
SLAM Example:

- Estimate the robot's poses and the landmarks at the same time



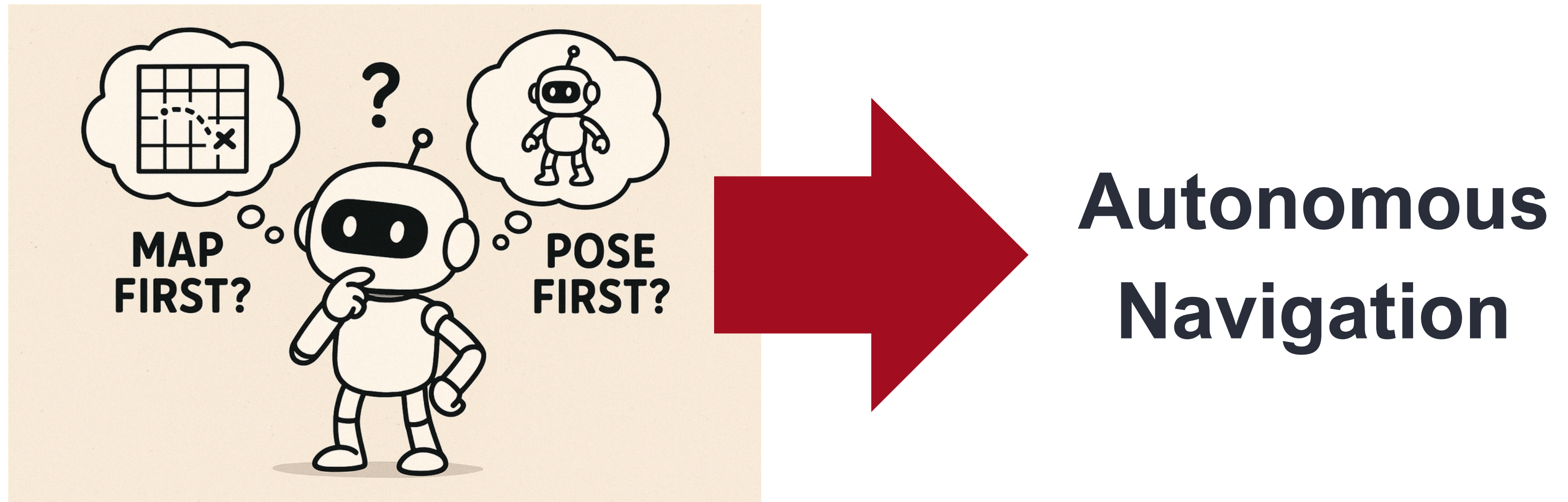
The SLAM Problem

- Slam is a **chicken-or-egg** problem:
 - A map is needed for localization
 - A pose estimate is needed for mapping



SLAM Is Relevant

- A fundamental problem for truly autonomous robots
- The basis for most navigation systems



SLAM Applications



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SLAM Applications

- Slam is central to range of indoor, outdoor, air, underwater applications for both manned and autonomous vehicles

Examples:

- **At home:** vacuum cleaner, lawn mower
- **Air:** surveillance with unmanned air vehicles
- **Underwater:** reef monitoring
- **Underground:** exploration of mines
- **Space:** terrain mapping for localization

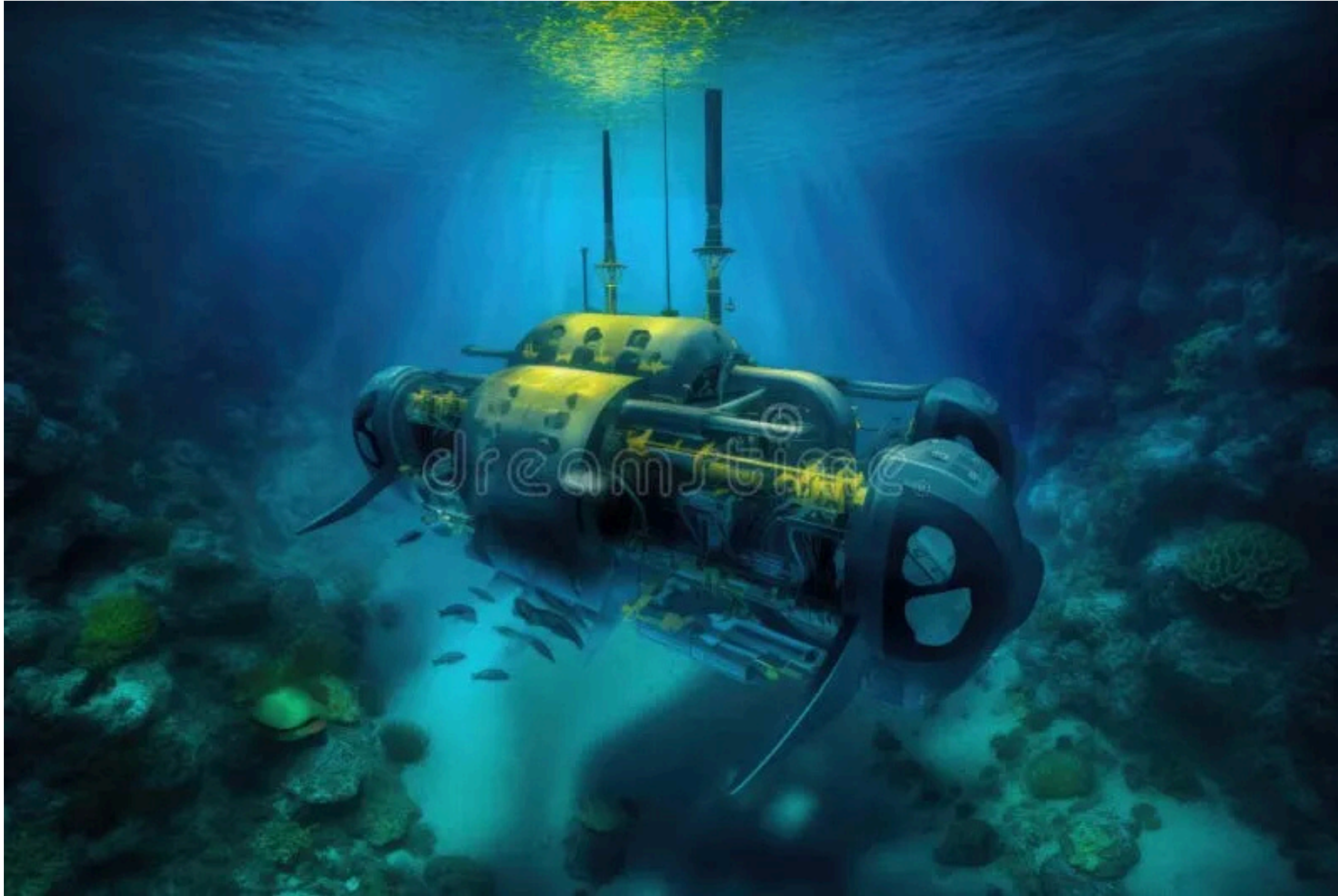


SLAM Applications

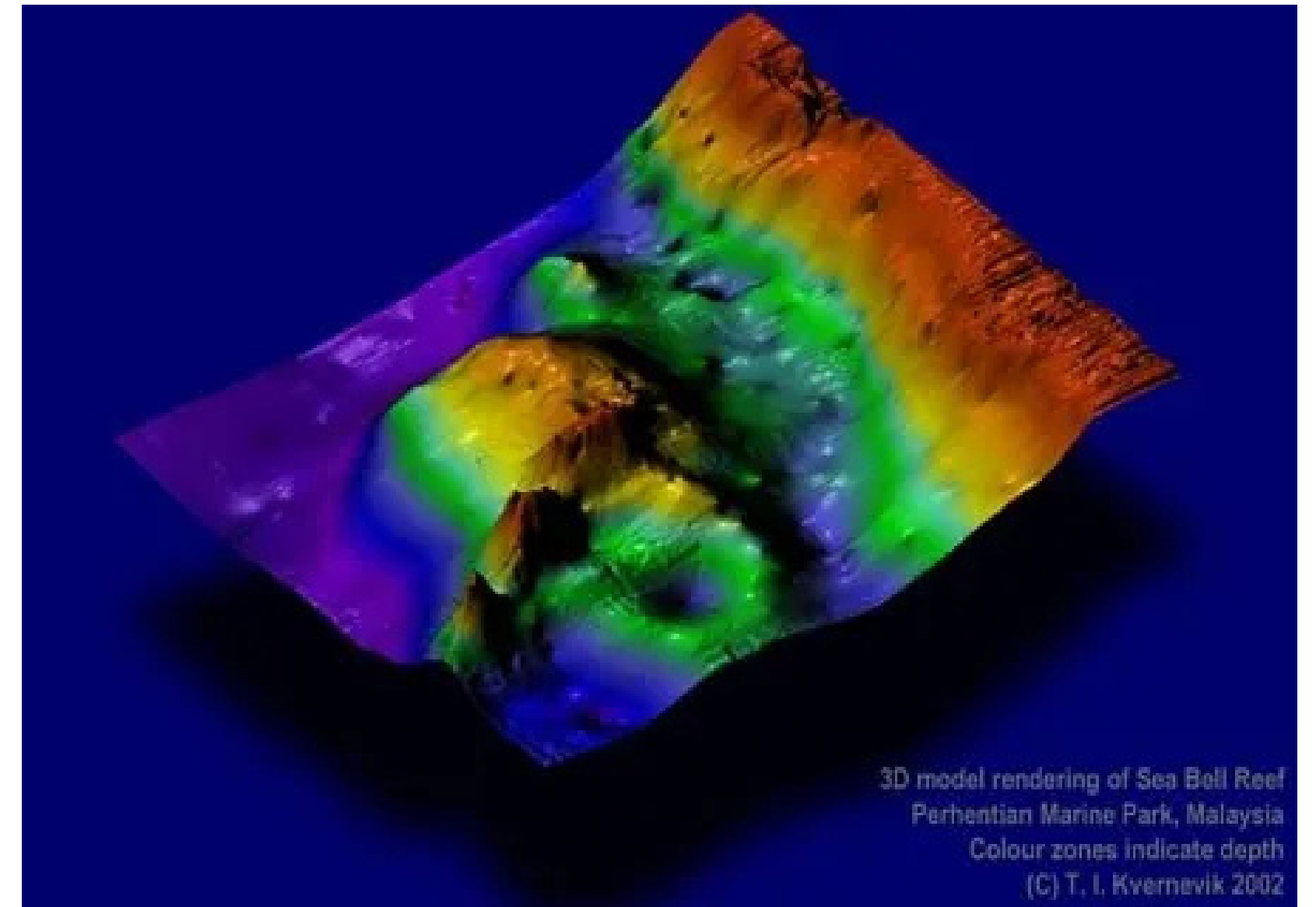
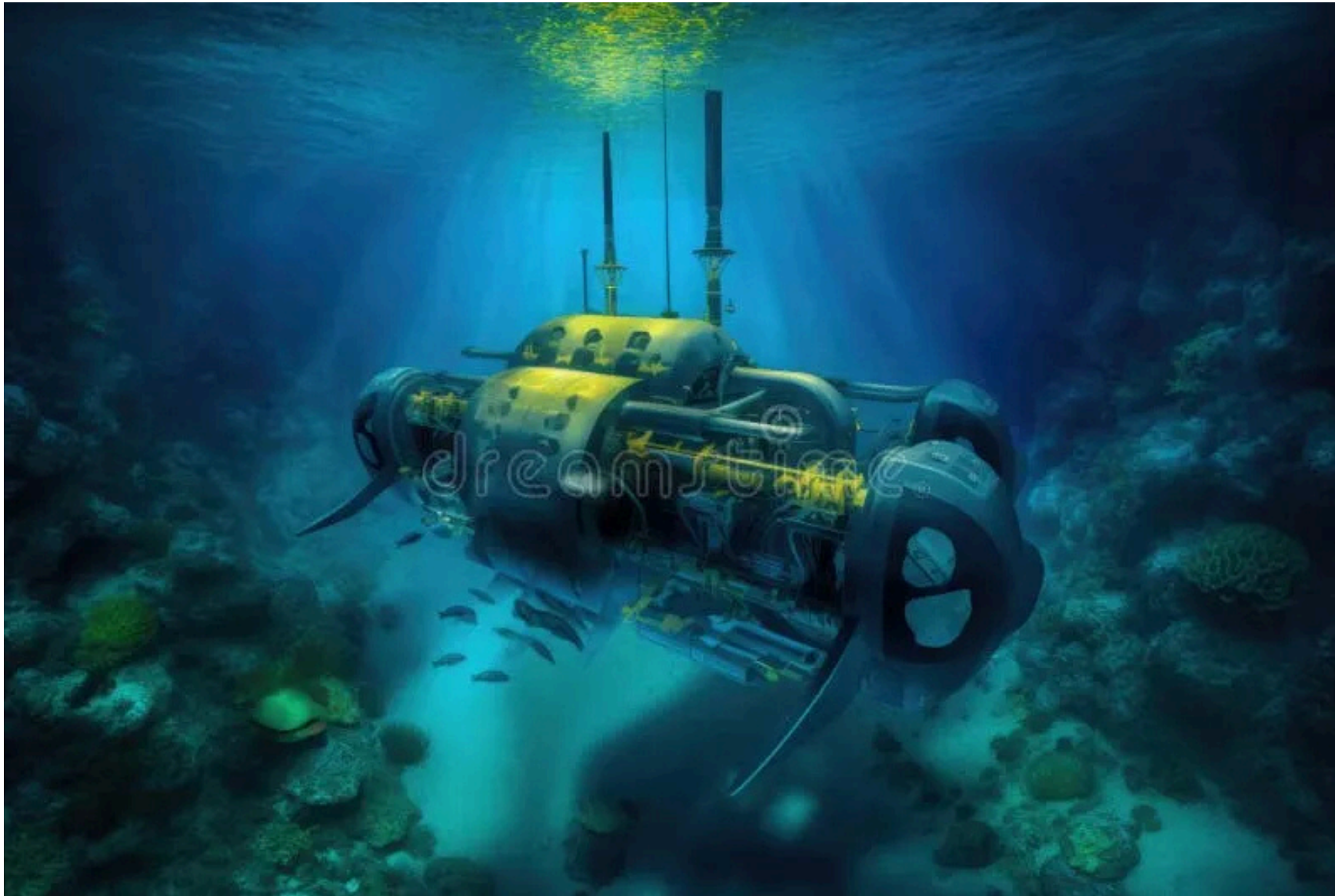


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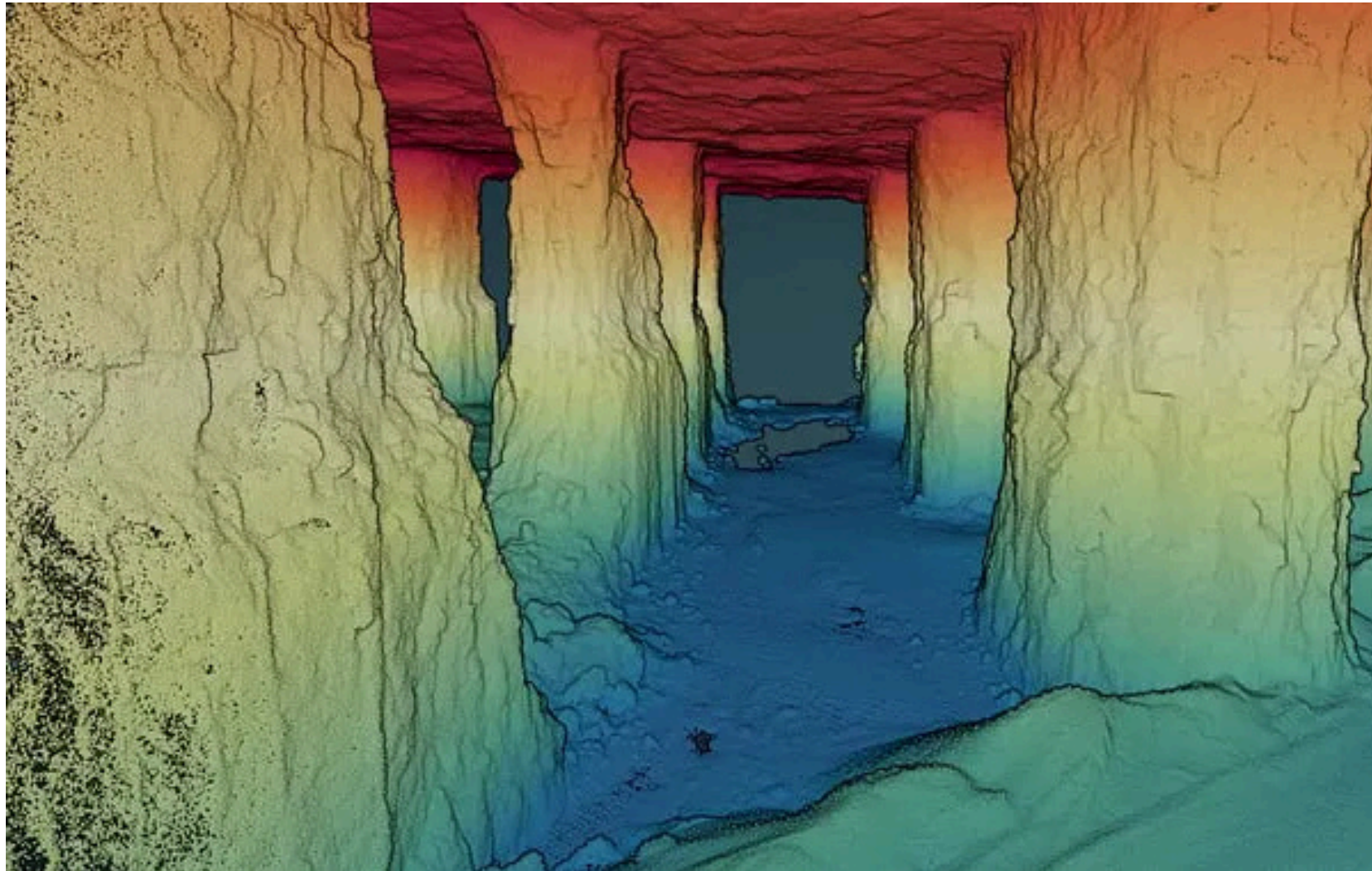
SLAM Applications



SLAM Applications

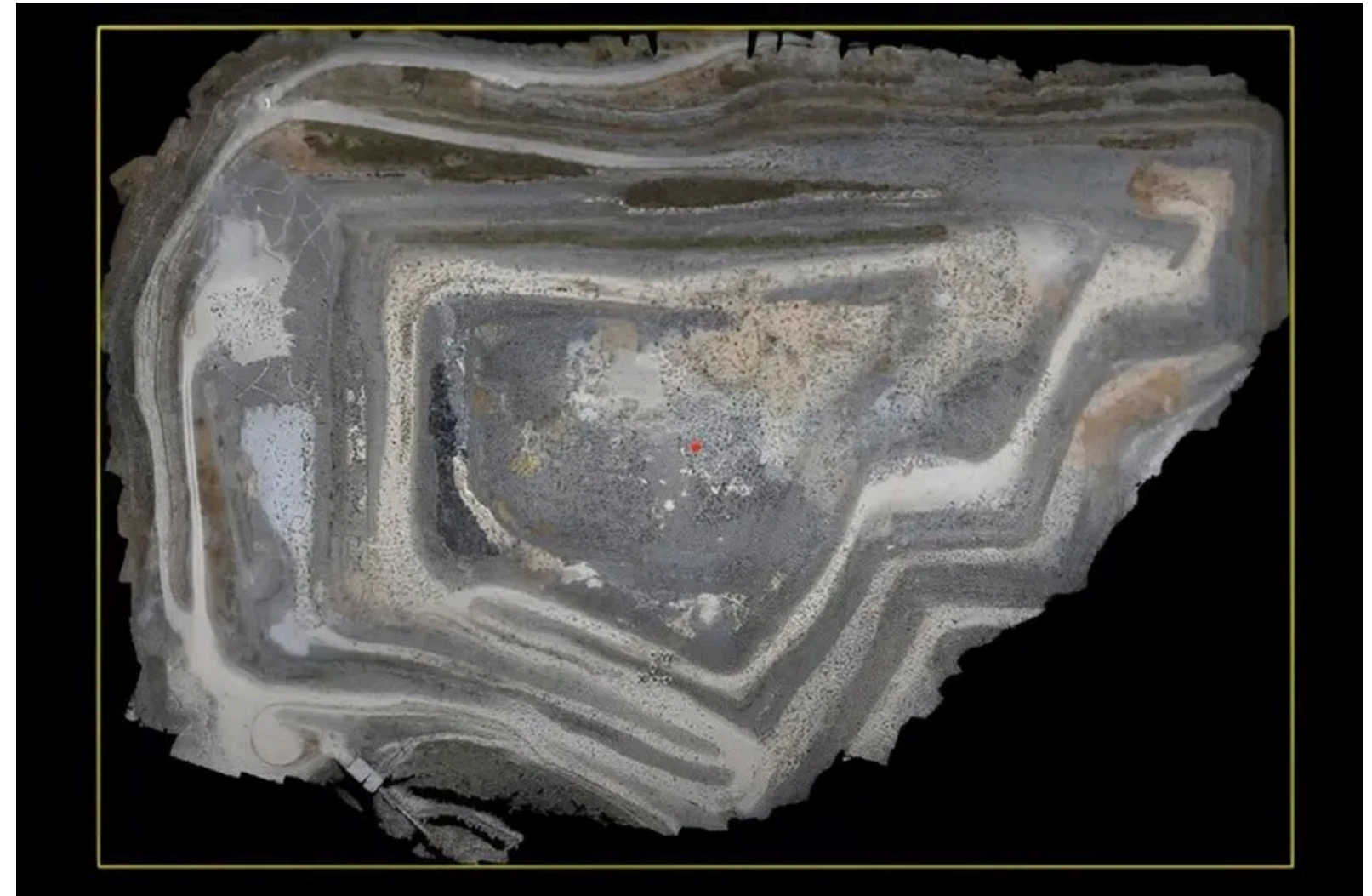


SLAM Applications



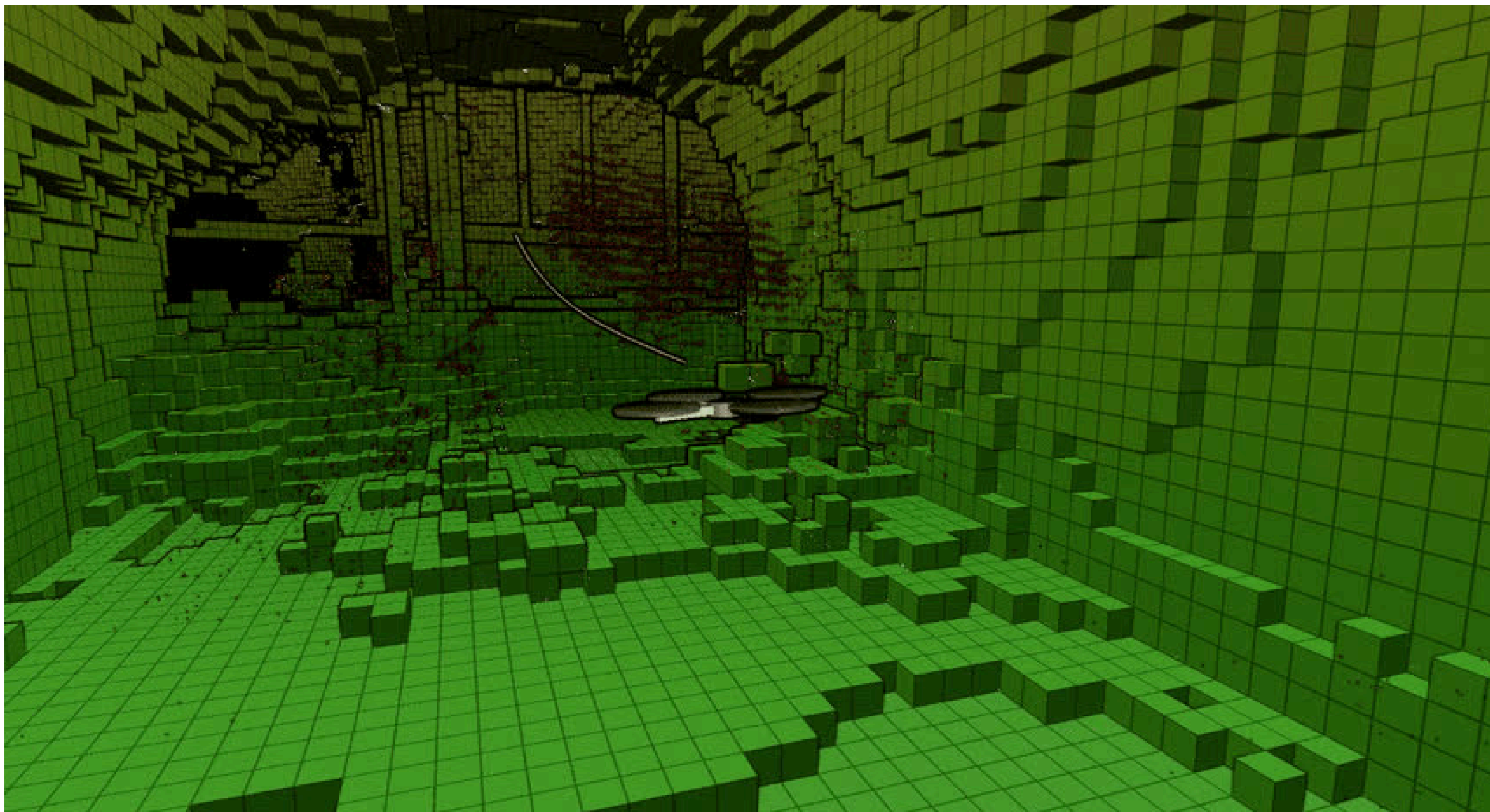
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SLAM Applications

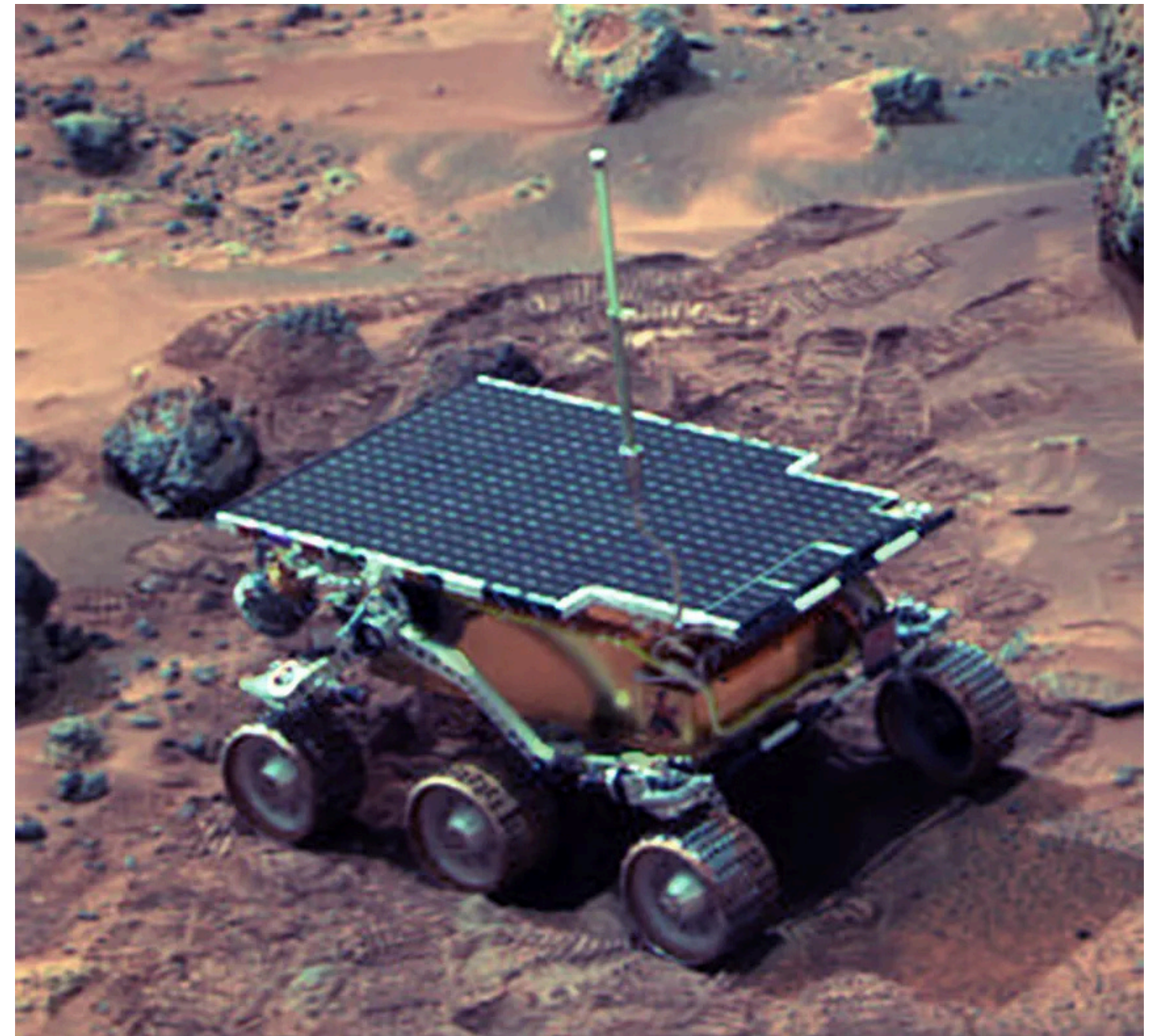
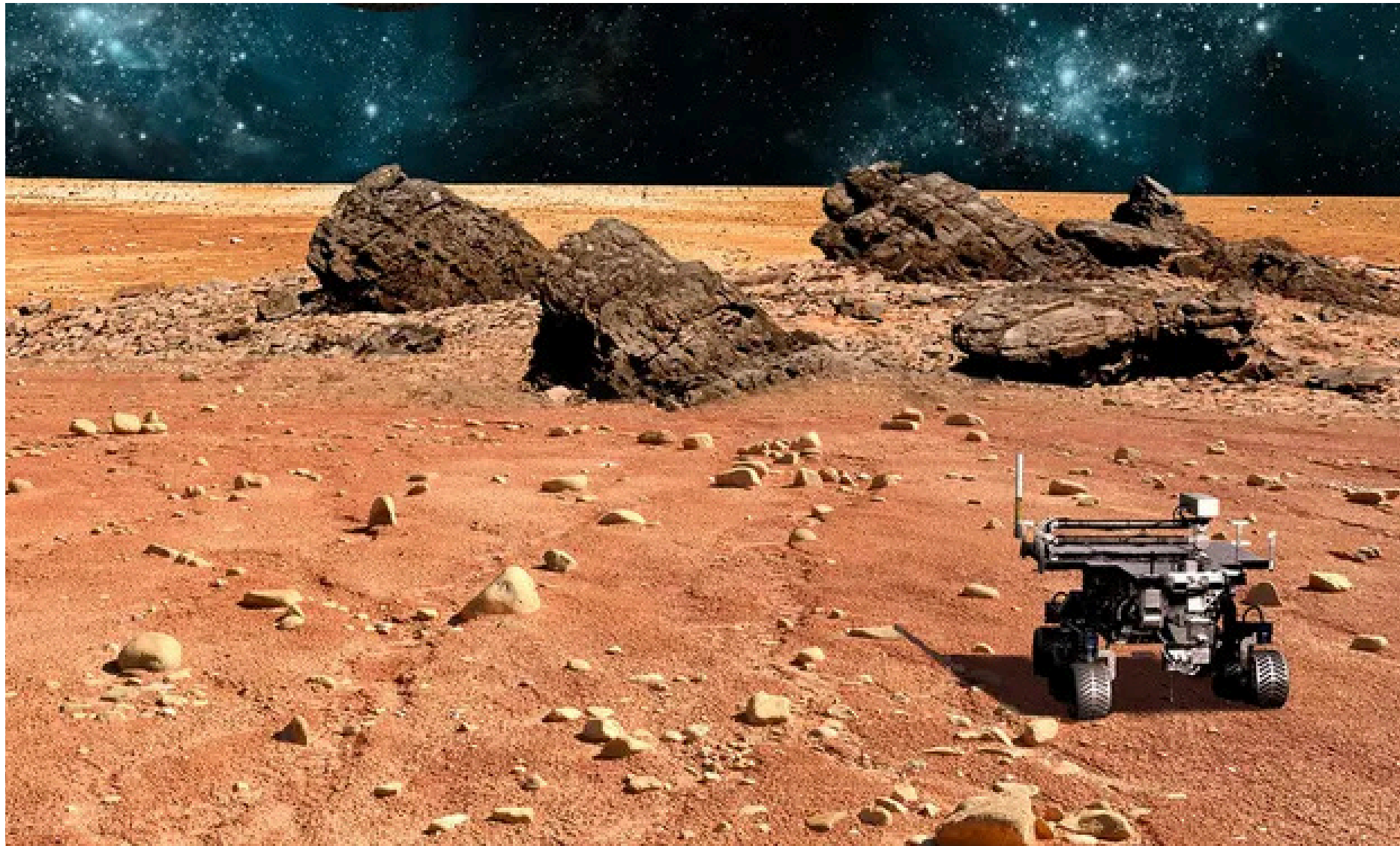


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SLAM Applications



SLAM Applications



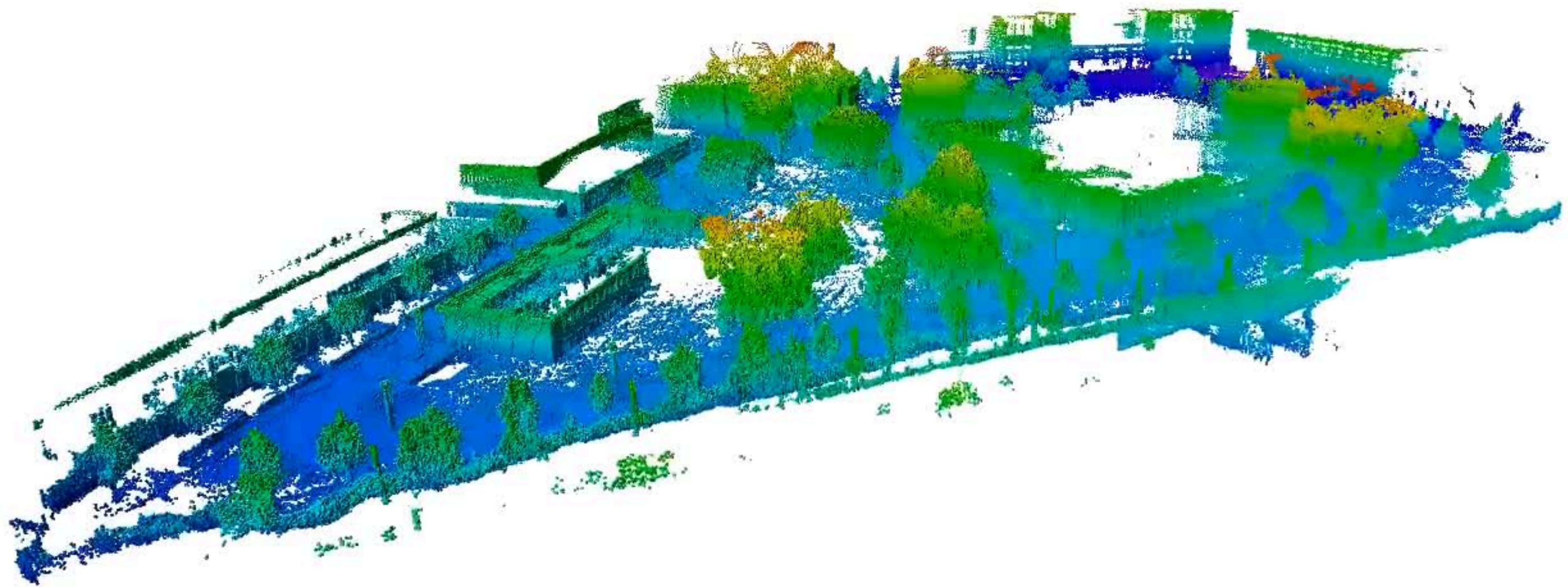
SLAM Applications



SLAM Applications

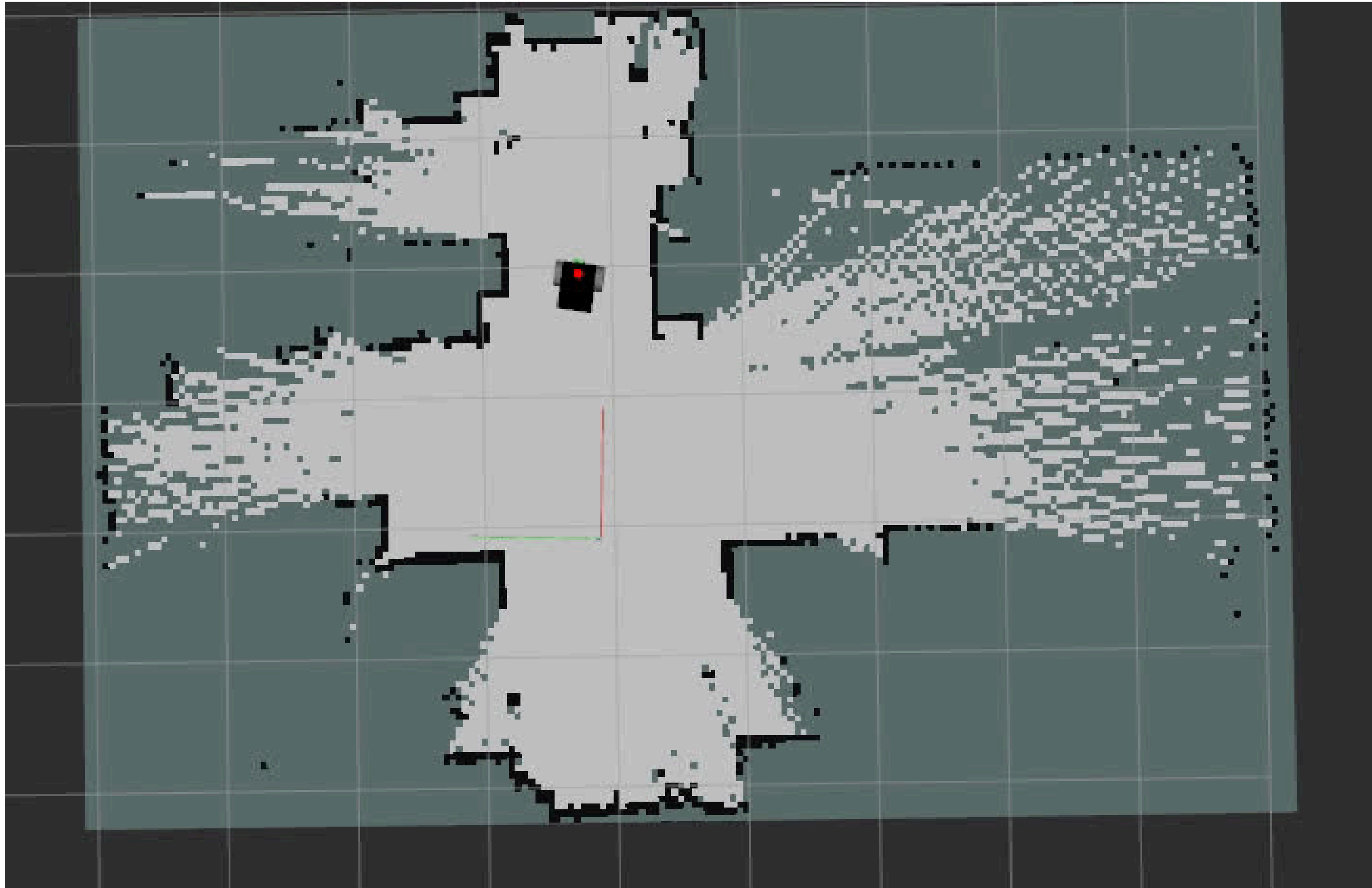


SLAM Applications



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SLAM Applications



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SLAM Problem



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SLAM Problem

Given:

- The robot's controls

$$u_{1:T} = \{u_1, u_2, u_3, \dots, u_T\}$$

- Observations

$$z_{1:T} = \{z_1, z_2, z_3, \dots, z_T\}$$

Wanted:

- Map of the environment

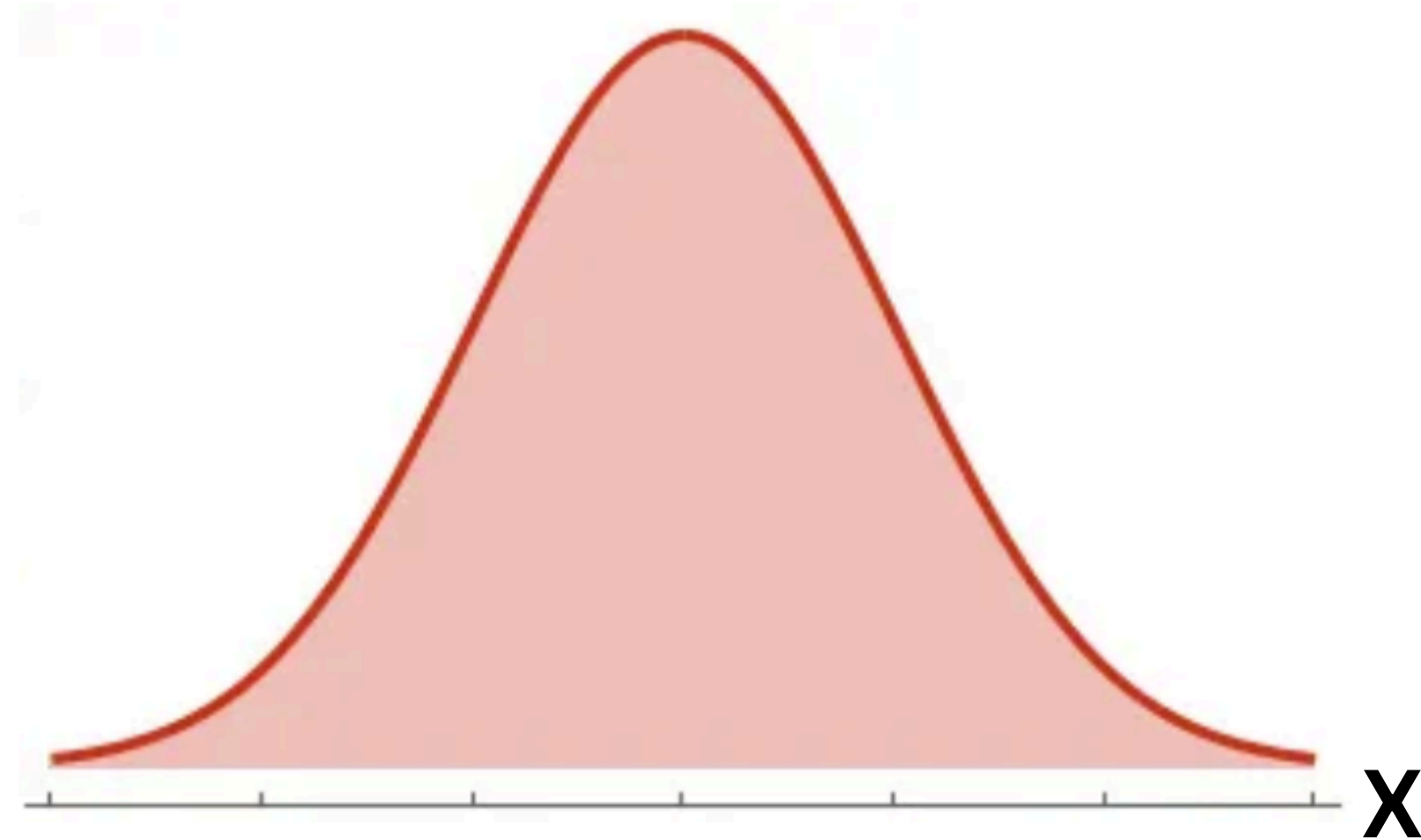
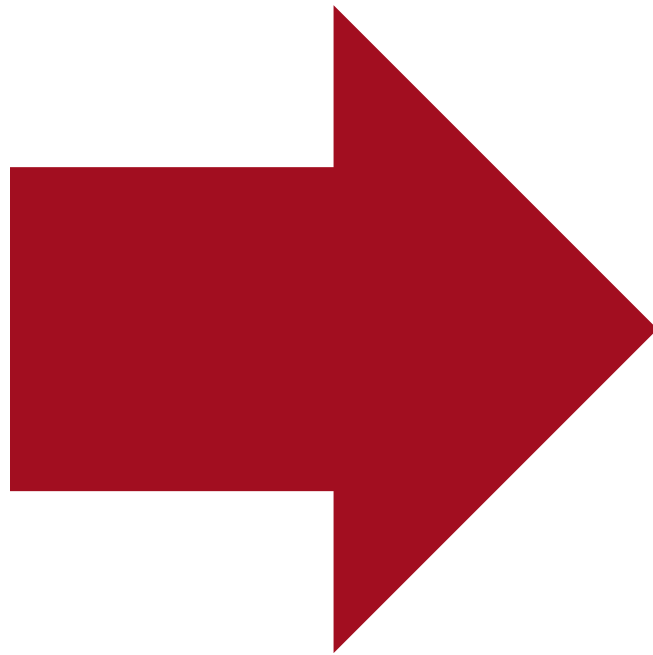
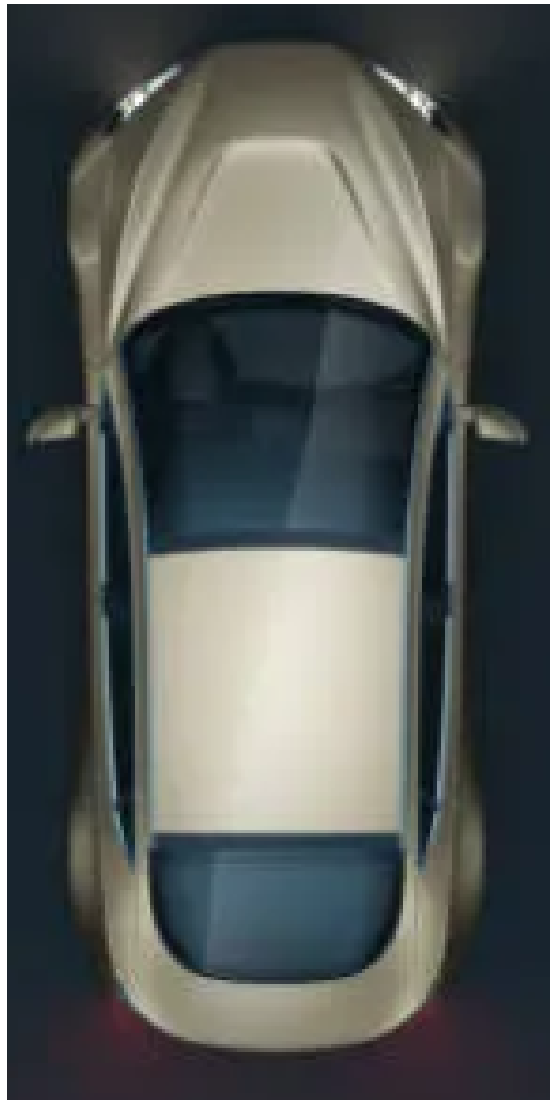
$$m$$

- Path of the robot

$$x_{0:T} = \{x_0, x_1, x_2, \dots, x_T\}$$

Probabilistic Approach

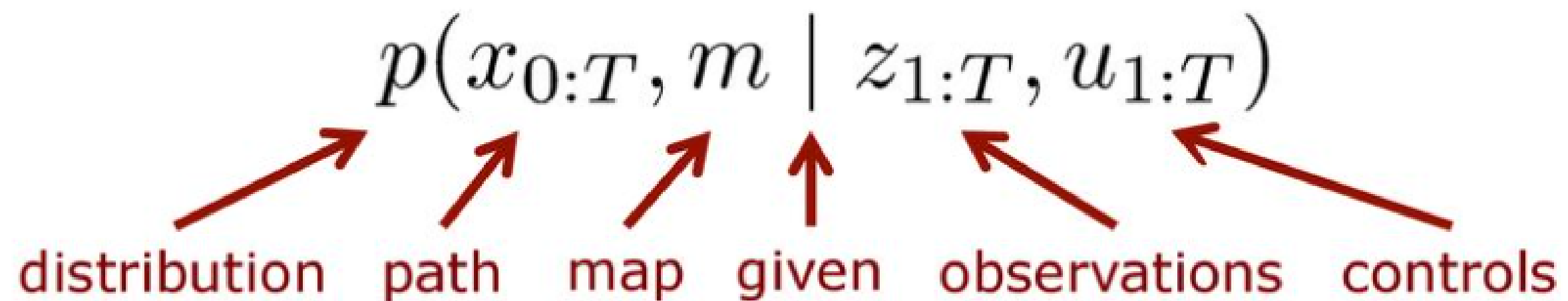
- Uncertainty in the robot's motions & observations
- Use the probability theory to explicitly represent the uncertainty



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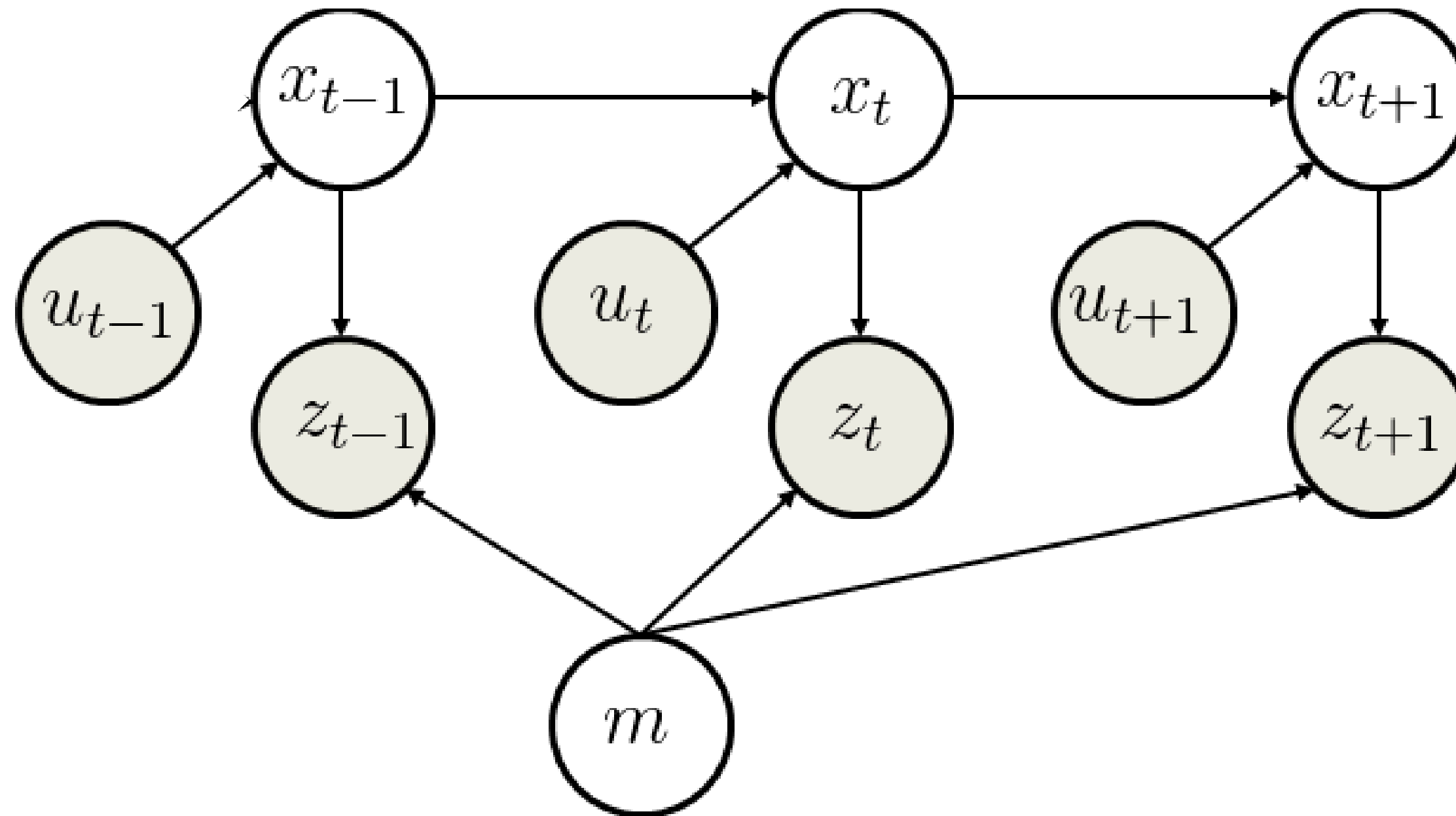
In The Probabilistic World

Estimate the robot's path and the map

$$p(x_{0:T}, m \mid z_{1:T}, u_{1:T})$$


distribution path map given observations controls

Graphical Model



Types of SLAM



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Types of SLAM

- **Full SLAM** – estimates the entire path

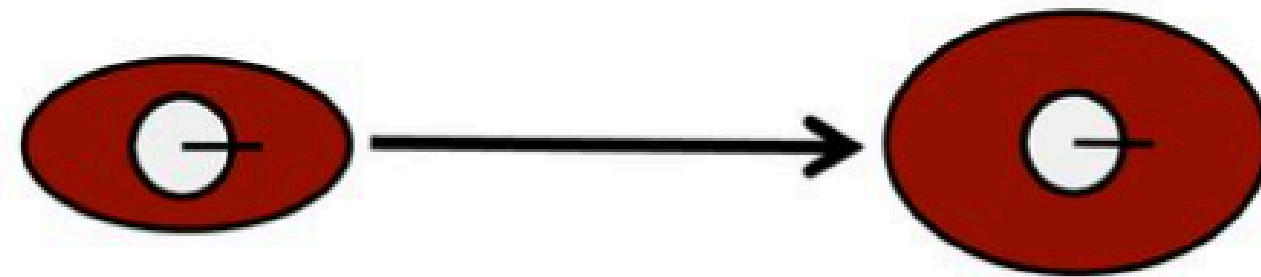
$$p(x_{0:T}, m \mid z_{1:T}, u_{1:T})$$

- **Online SLAM** – seeks to recover only the most recent pose

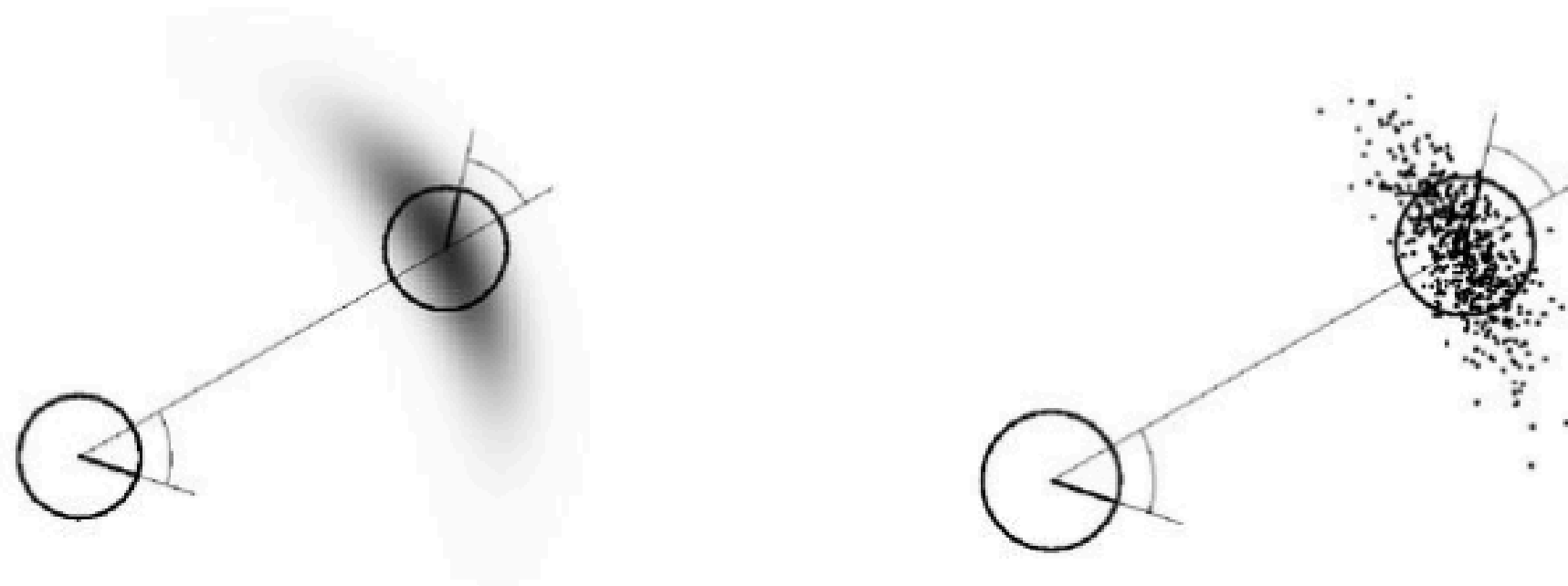
$$p(x_t, m \mid z_{1:t}, u_{1:t})$$

Motion Model Examples

- Gaussian model

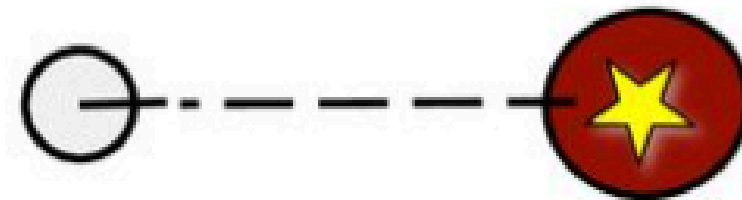


- Non-Gaussian model

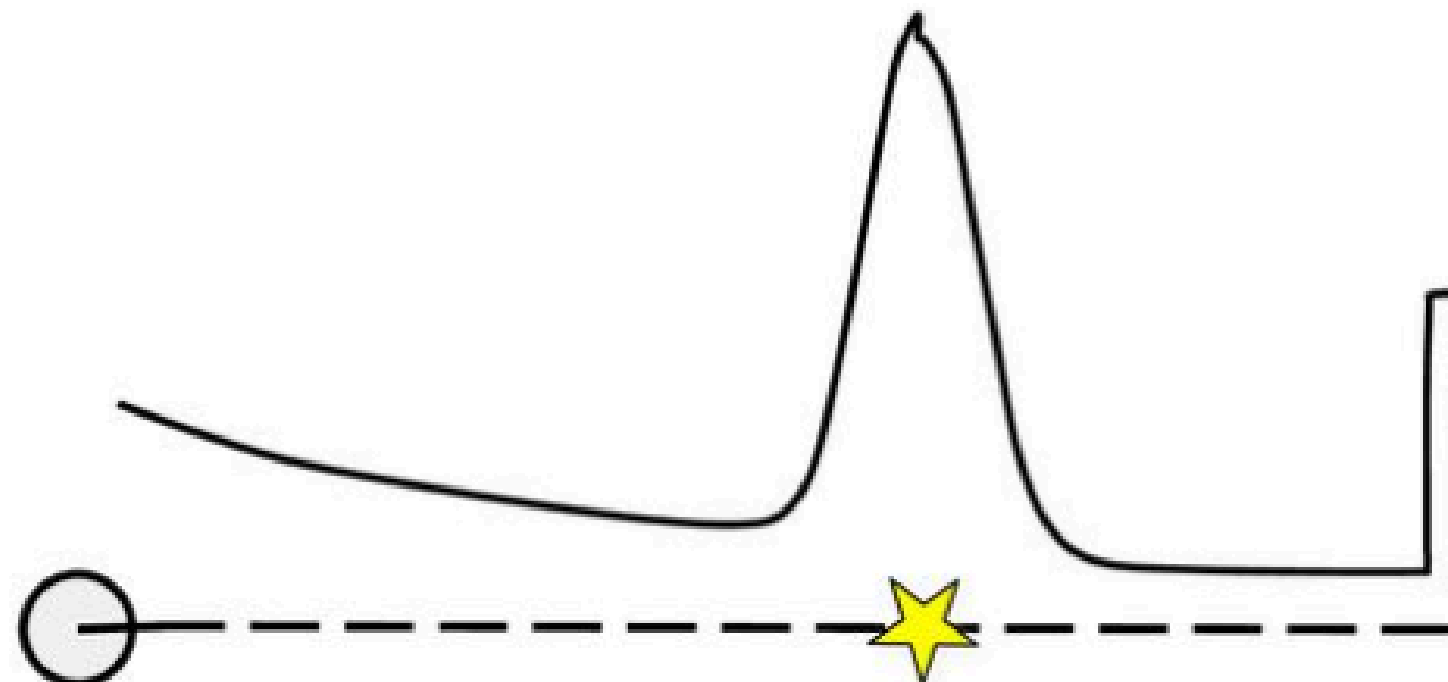


Observation Model Examples

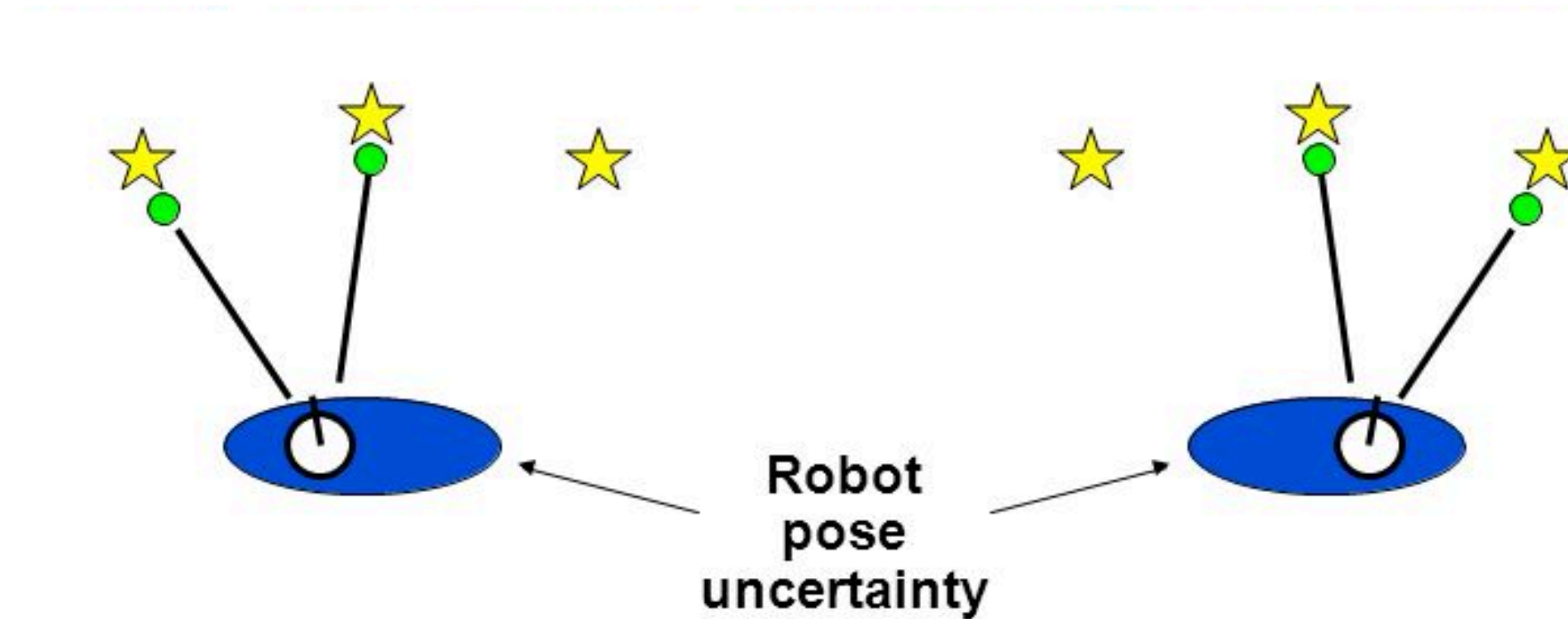
- Gaussian model



- Non-Gaussian model



Why is SLAM a hard problem ?



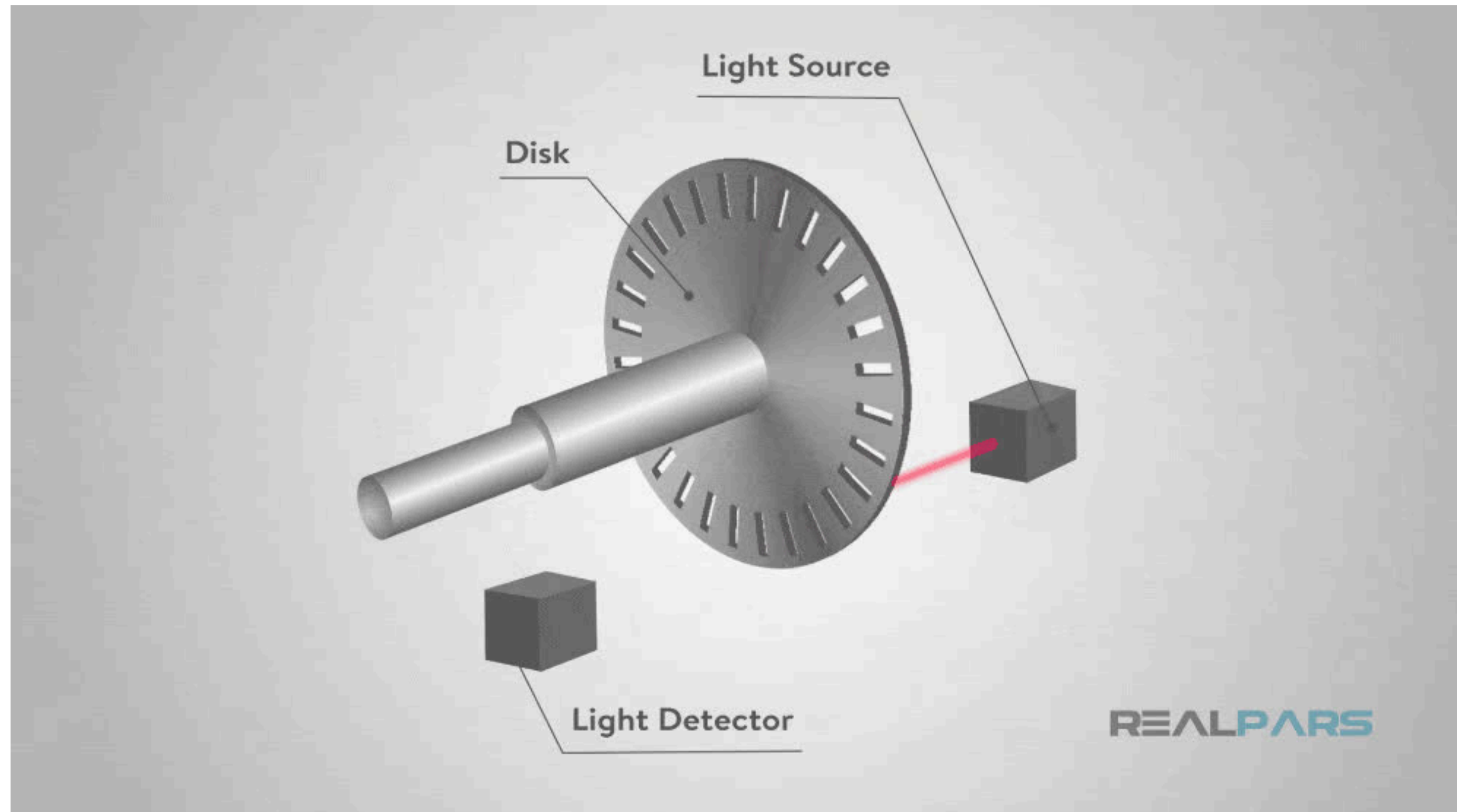
- In the real world, the mapping between observations and landmarks is unknown
- Picking wrong data associations can have catastrophic consequences
- Pose error correlates data associations

Sensors

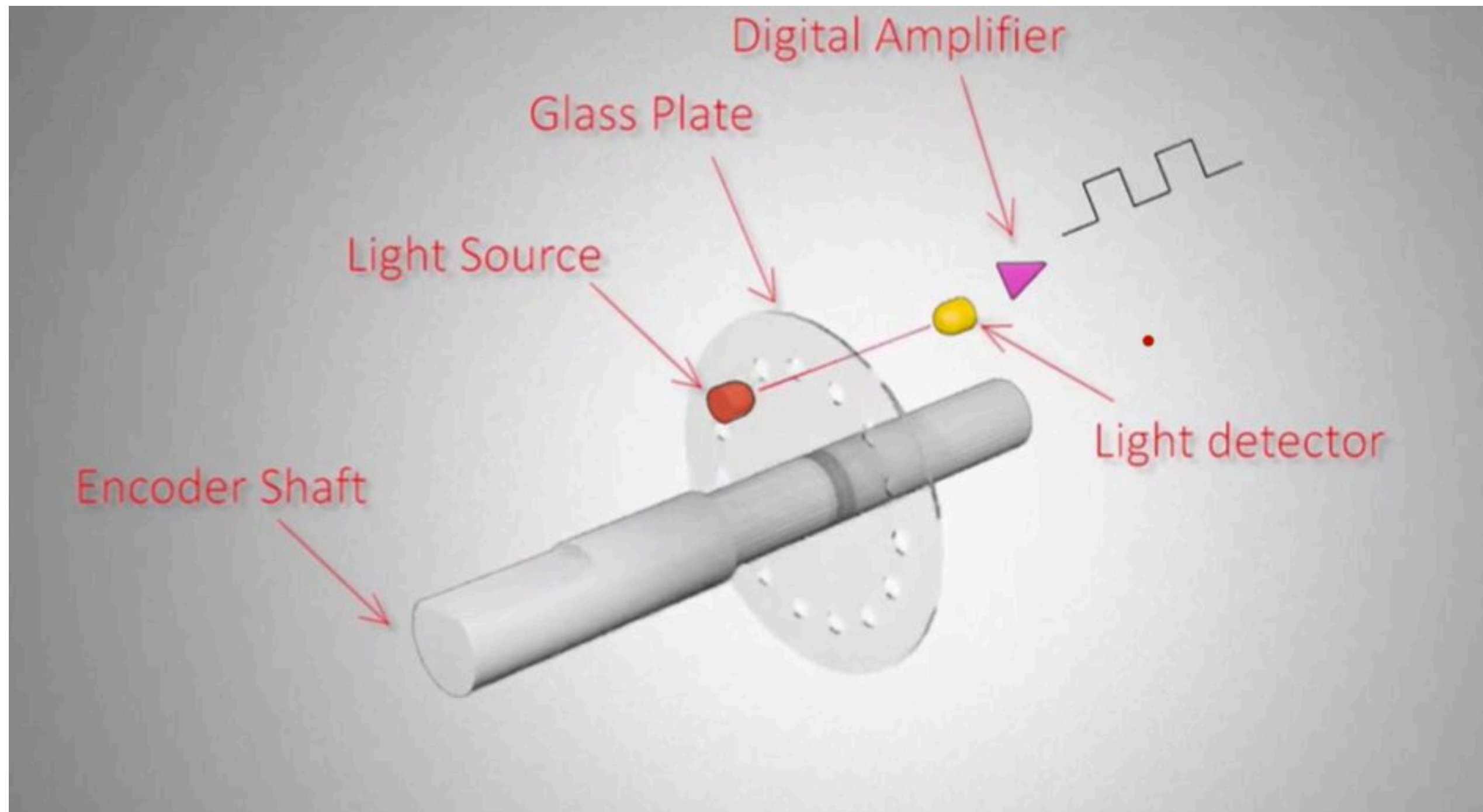


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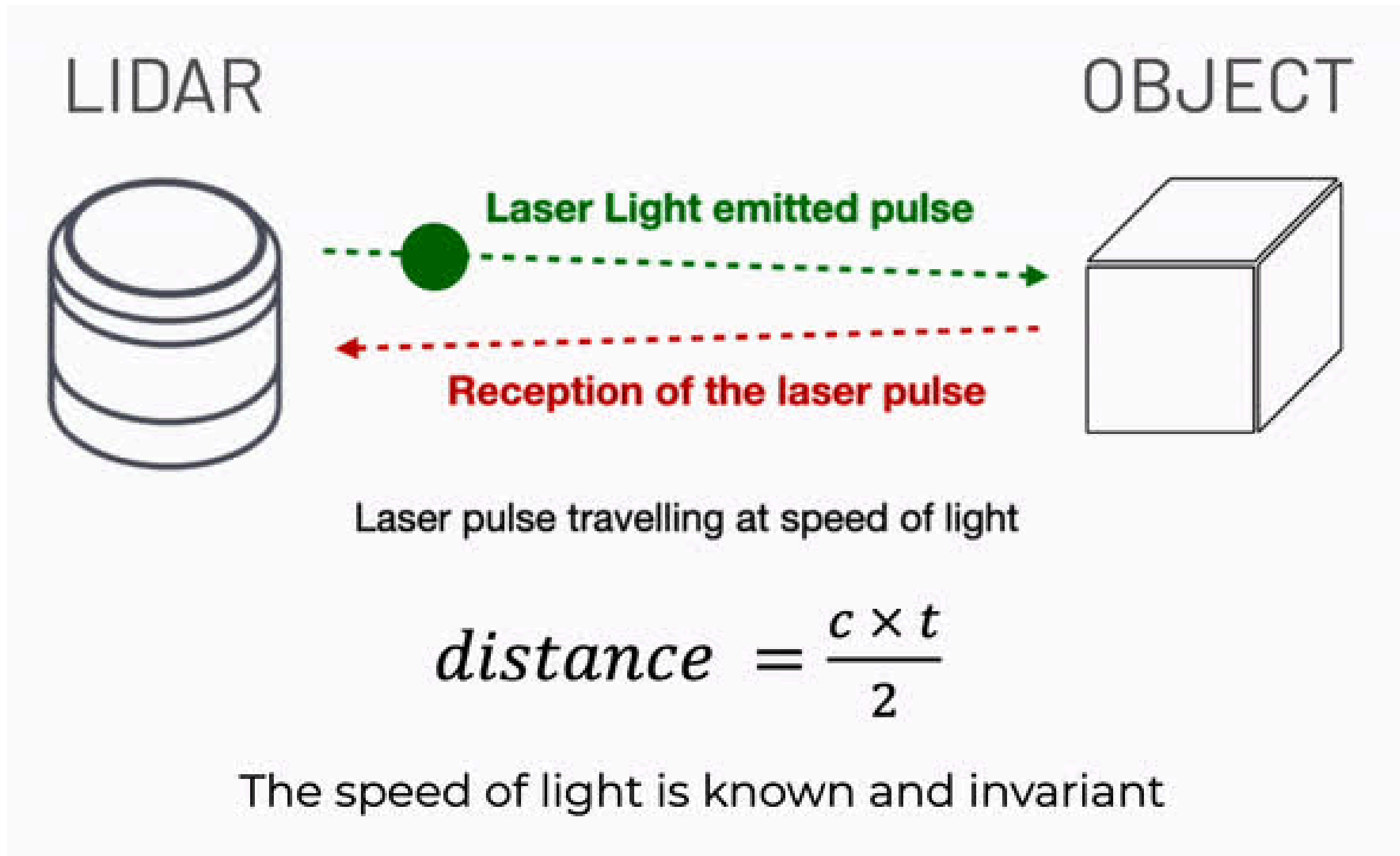
Encoder



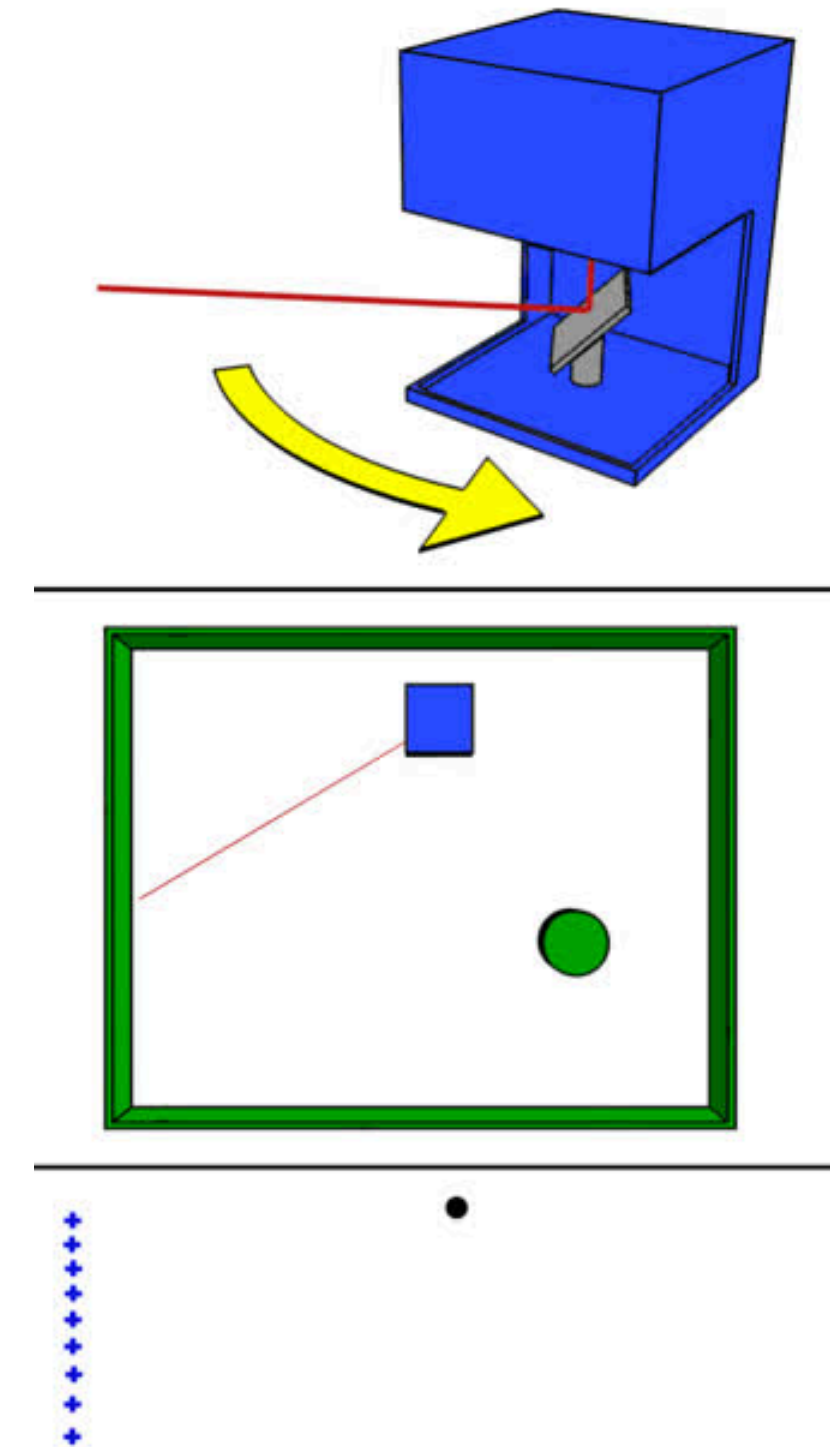
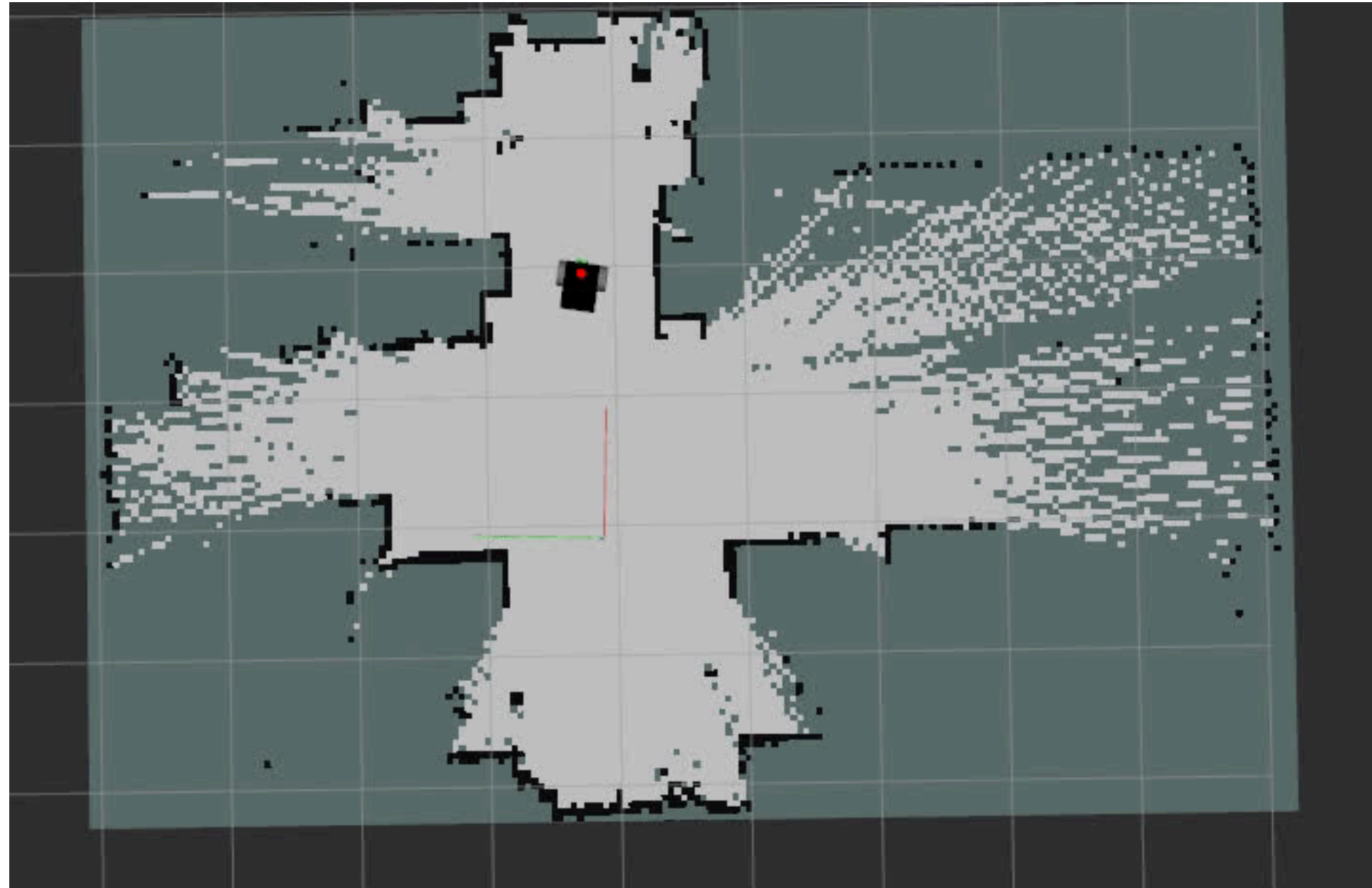
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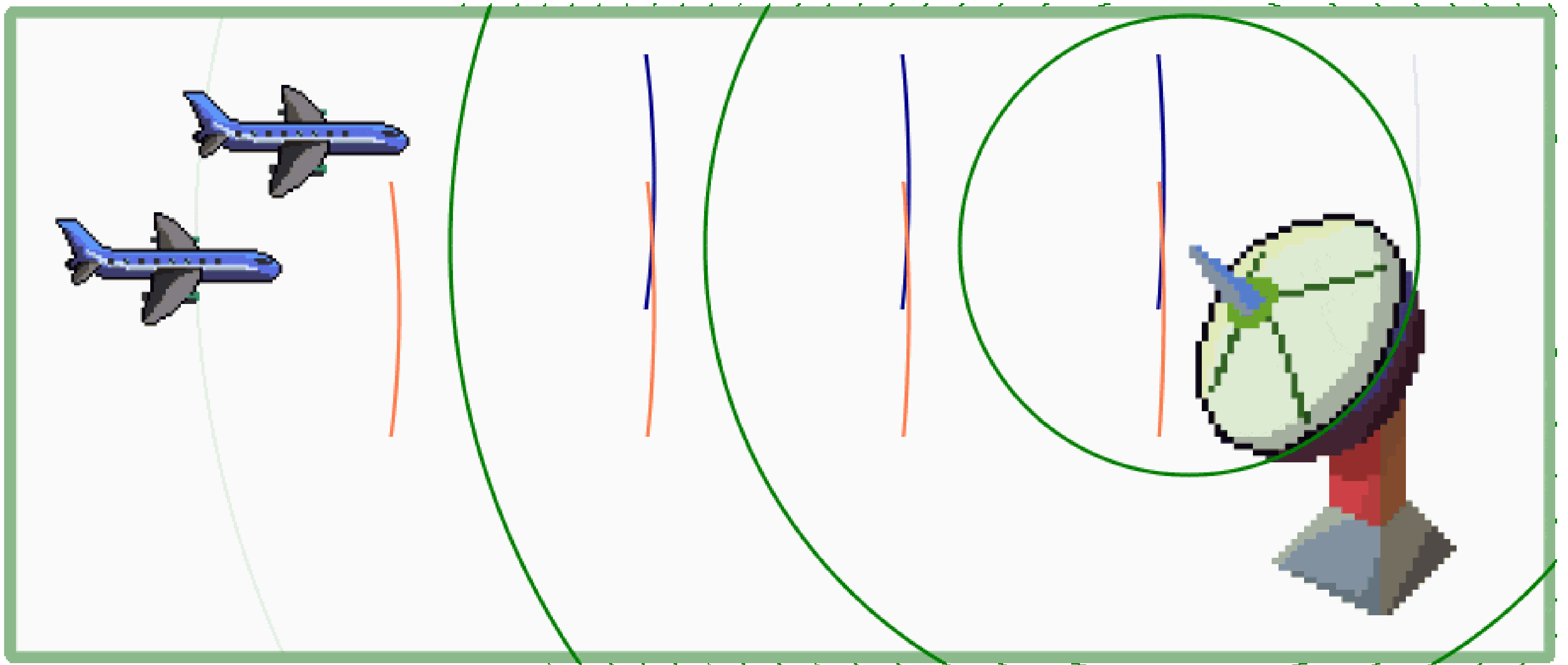
LIDAR



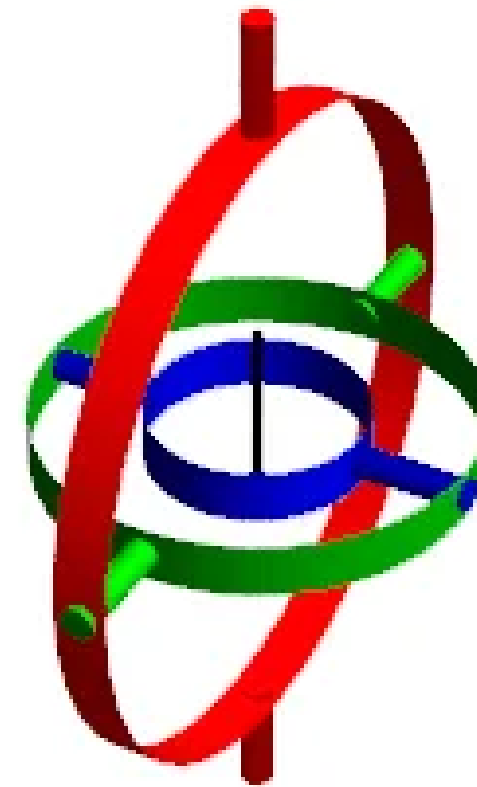
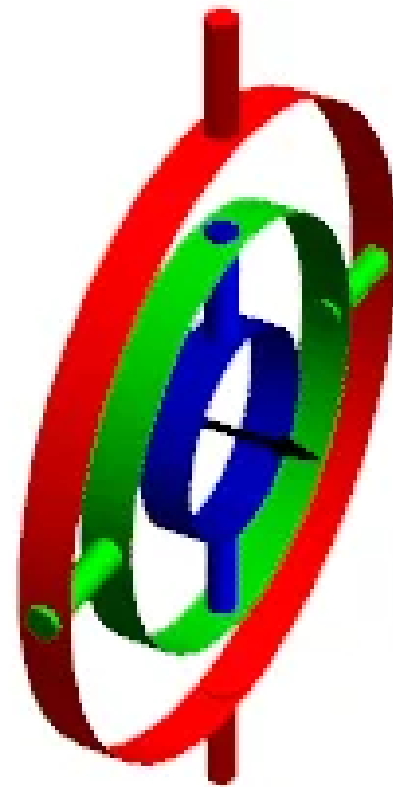
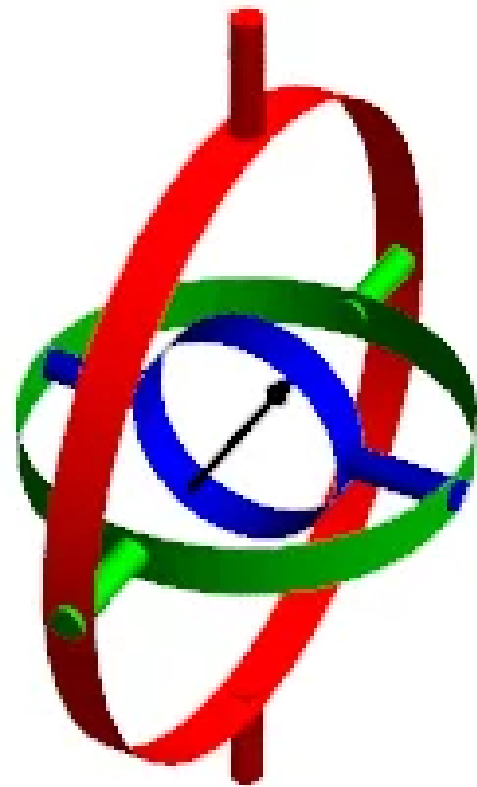
LIDAR



RADAR



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Thank You



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